Yuchen Zhou

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EDUCATION

Carnegie Mellon University Pittsburgh, PA
Master of Science in Information Networking | GPA: 3.86/4.0 Aug. 2025 – Dec. 2026

Nankai University Tianjin, China
Bachelor of Engineering in Information Security | GPA: 3.71/4.0 Sep. 2021 – Jul. 2025

SKILLS

Languages: C++, Python, Java, Go, C, Rust, SQL (PostgreSQL, MySQL), NoSQL (Neo4j, DynamoDB), Shell, HCL
System & Backend: Git, OOP, System Design, CI/CD, Linux/Unix, Redis, Kafka, Docker, Kubernetes, MapReduce, Hadoop, Terraform, REST APIs, FastAPI, Database Internals, Distributed Systems, Microservices, Query Optimization
AI Infra & Cloud: AWS (Bedrock, S3, ECS, DynamoDB, EventBridge, AOSS), GCP, LLM, NLP, MCP, PyTorch, Hugging Face, Transformers, ReAct, Multi-Agent Systems, CUDA, Speculative Decoding, Claude Code, Codex

PROFESSIONAL EXPERIENCE

ENFOS Inc. Durham, NC
Software Engineering Intern May 2026 – Aug. 2026

- Designed and delivered a scalable ReAct-based multi-agent architecture for 100+ tools via a tool-search layer, reducing average tool-calling latency by 18% while improving tool-selection accuracy by 40%.
- Consolidated 5+ MCP-based agent context mechanisms into hybrid BM25/semantic retrieval over messages, answers, and attachments for long-term memory and a verbatim short-term window, cutting average context-token usage by 61% while achieving 79% Recall@5 across 100+ turn conversations.
- Architected and provisioned AWS-based agent infrastructure using EventBridge, SQS, Lambda, DynamoDB, Bedrock, and AOSS for support async attachment ingestion, vector retrieval, and long-term agentic memory.
- Built an async and scalable document lifecycle (upload/update/delete/bulk import/set secure) ingestion and indexing of 500M documents across 100K sites, handling over 20+ different edge cases and failure scenarios.

Chinasoft International Beijing, China
Backend Software Engineering Intern May 2023 – Jul. 2023

- Led backend Hive/HDFS-based MapReduce workflow over 4M+ records, boosting query performance by 32%.
- Delivered Spring Boot backend services and REST APIs for interactive data analytics and virtualization while coordinating closely with frontend engineers and non-technical designers to ship product features.
- Engineered MySQL data ingestion and streaming systems using Sqoop, Kafka, and ZooKeeper, ensuring reliable backend data flow and reducing end-to-end processing latency by 27%.

ACADEMIC PROJECTS

Relational Database Systems – BusTub DBMS (GitHub) Feb. 2026 – May 2026

- Implemented core DBMS internal components in C++, including a disk-oriented buffer pool manager, B+ tree indexing, transaction-aware executors for concurrency, batch-based query execution engine, and optimization rules.
- Extended BusTub's rule-based optimizer with 5+ plan rewrite rules, including filter-scan merging, SeqScan and OrderBy to IndexScan selection, NestedLoopJoin to HashJoin, Sort-Limit-to-TopN optimization.
- Built operators (e.g., external merge sort) improving benchmark by 56% and ranking in the top 5% leaderboard.

AI Infra: LLM Inference, Distributed Training & GPU Kernels (GitHub) Feb. 2026 – May 2026

- Integrated draft-model-free suffix speculation into Hugging Face Transformers, reusing context patterns to achieve 1.74x decoding speedup on the WildChat code-suffix benchmark. (GitHub)
- Designed a unified verification interface for suffix and tree speculation, enabling multiple candidate continuations to be verified in one target-model pass and achieving 1.30x speedup on SWE-bench tree-spec benchmark.
- Implemented data and Megatron-style tensor parallelism using MPI collectives, with ZeRO-3-style sharding of parameters, gradients, and optimizer states to reduce per-worker memory usage.
- Optimized FP16 GEMM kernels for NVIDIA Blackwell using TVM/TIRX, combining async TMA loads, Tensor Core operations, software pipelining, and warp specialization to overlap data movement with computation.